

AI-Powered E-commerce Funnel & Conversion Intelligence

Project Overview

Modern e-commerce businesses generate large volumes of behavioral data, but converting this data into actionable business insights remains a challenge. The objective of this project was to build an end-to-end analytics platform capable of identifying user behavior patterns, measuring conversion performance, and generating AI-powered business insights.

The platform analyzes customer journeys from initial website visits to purchase completion, identifies drop-off points in the conversion funnel, evaluates traffic source quality, and provides automated recommendations using Large Language Models (LLMs).

The final solution was deployed as a live web application using Flask and Render.

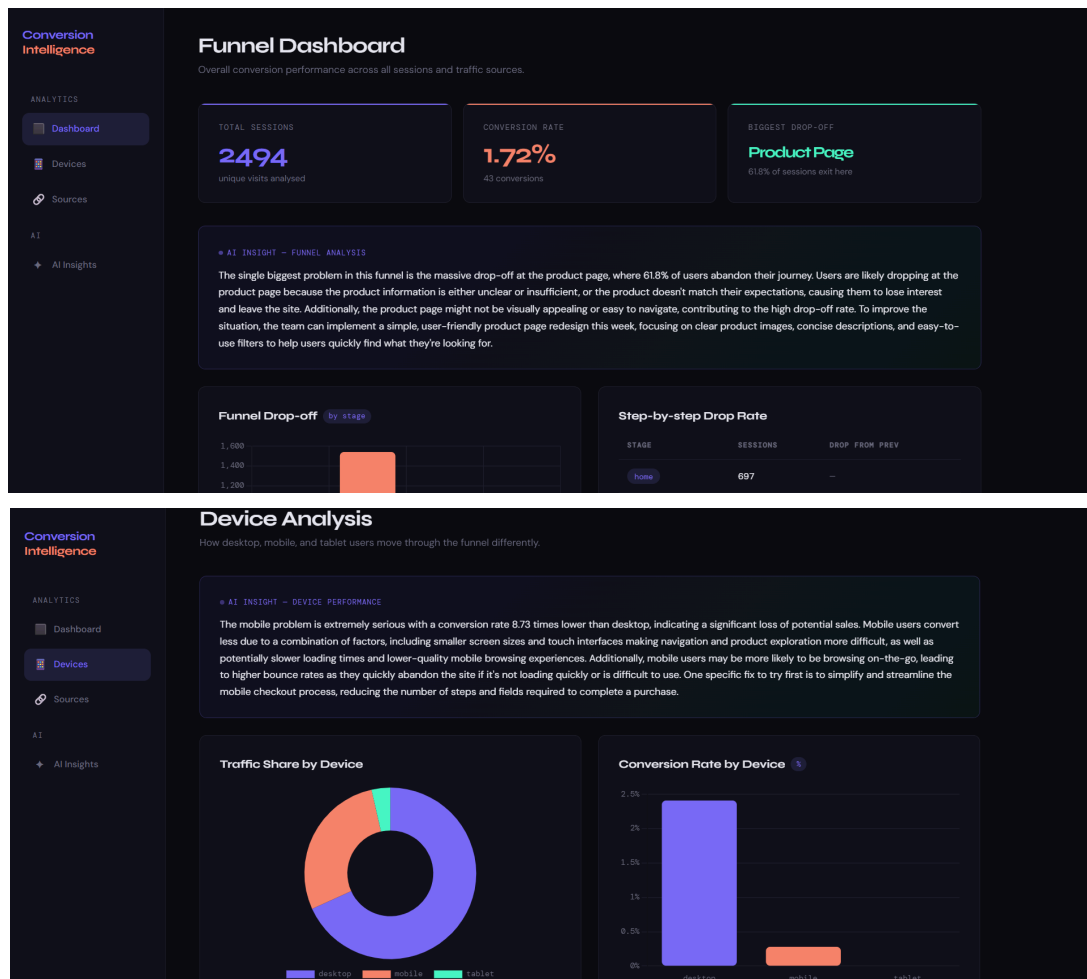
Business Problem

E-commerce teams often struggle to answer questions such as:

- Where do customers abandon their purchase journey?
- Which marketing channels bring high-quality traffic?
- Do mobile users behave differently from desktop users?
- Which pages contribute most to conversion success or failure?
- How can these insights be communicated to non-technical stakeholders?

Traditional dashboards display metrics but require manual interpretation. This project combines analytics with AI-generated insights to reduce the gap between data and decision-making.

Dashboard Visuals:



Dataset

Source: Google Analytics E-commerce sample dataset(Big Query)

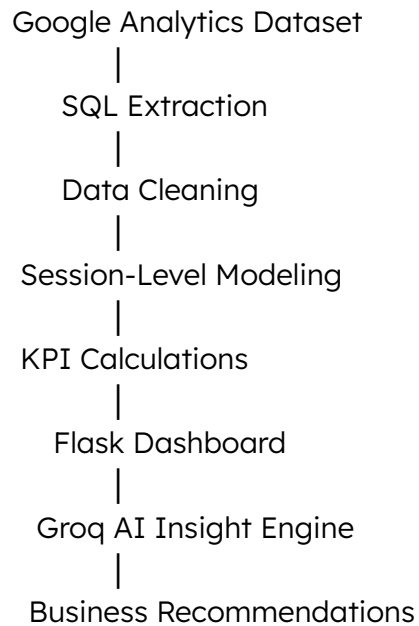
Data Collected

The dataset contains:

- Session IDs
- Page URLs
- Device types
- Traffic sources
- Transaction information
- User navigation behavior

The data was extracted using SQL queries and transformed into session-level analytics tables.

Solution Architecture



Data Processing Pipeline

Step 1: Data Extraction

Relevant fields were extracted from the Google Analytics dataset using SQL.

Collected attributes included:

- Session identifiers
- Page visits
- Traffic sources
- Device categories
- Transactions

Step 2: Page Classification

Each page was mapped into a funnel stage.

Page Type	Funnel Category
Home Page	Home
Product Pages	Product
Cart Pages	Cart
Checkout Pages	Checkout
Others	Other

This classification allowed customer journeys to be analyzed at a business level rather than at individual URL level.

Step 3: Session Funnel Construction

User journeys were reconstructed session by session.

Example:

Home



Product



Cart



Checkout

Each session was assigned:

- Funnel Stage
- Last Step Reached
- Conversion Status
- Drop-off Category

The funnel stage is based on the last staged reached i.e if a user adds a product to cart but doesn't checkout, the last stage reached is Cart and a funnel stage is assigned accordingly.

Step 4: Traffic Source Categorization

Raw traffic sources were grouped into:

- Google / Paid
- Direct
- YouTube
- Partners
- Other

This enabled source-level performance analysis.

Key Metrics

- Conversion Rate

Measures the percentage of sessions that completed a purchase.

Business Value

Indicates the effectiveness of the website in turning visitors into customers.

- Funnel Drop-off

Tracks where users abandon the purchase journey.

Business Value

Highlights friction points that prevent conversions.

- Bounce Rate

Percentage of users leaving after minimal engagement.

Business Value

Evaluates landing page effectiveness and traffic quality.

- Device Performance

Compares behavior across:

1. Mobile

2. Desktop
3. Tablet

Business Value

Identifies user experience issues across devices.

- Source Performance

Measures conversion performance by traffic source.

Business Value

Helps marketing teams allocate budget effectively.

Dashboard Features

Executive Dashboard

Displays:

- Total Sessions
- Conversion Rate
- Funnel Visualization
- AI-generated Insights

Device Analytics

Provides:

- Device Share
- Device Conversion Rate
- Device-specific Drop-off Analysis

Traffic Source Analytics

Displays:

- Source Performance
- Bounce Rate by Source
- Conversion Quality

AI Insight Engine

The platform integrates Groq LLMs to generate business-friendly explanations from aggregated metrics.

Instead of displaying raw numbers, the system explains:

- Major drop-off points
- Conversion opportunities
- Source quality issues
- Device-specific challenges

This makes the dashboard accessible to non-technical stakeholders.

Sample Business Findings

The analysis revealed several important patterns:

Funnel Friction

A significant percentage of users dropped off during the product exploration stage, suggesting that product discovery and merchandising could be improved.

Mobile Performance Gap

Mobile users exhibited lower conversion rates compared to desktop users, indicating potential usability issues on smaller screens.

Source Quality Differences

Certain traffic sources generated high session volumes but low conversion rates, suggesting lower traffic quality.

Checkout Abandonment

A noticeable proportion of users exited before completing checkout, highlighting an opportunity to optimize the purchasing process.

Technical Stack

Backend

- Python
- Flask

Data Processing

- Pandas
- SQL

AI Layer

- Groq API
- Prompt Engineering

Deployment

- Render

Version Control

- Git
- GitHub

Visualization

- Chart.js
- Custom Dashboard UI

Challenges Faced

1. Session-Level Funnel Modeling

The original dataset contained event-level records rather than complete customer journeys. Session reconstruction was required to build meaningful funnel analytics.

2. Deployment

The application was containerized and deployed to a cloud environment while securely managing API credentials through environment variables.

Results

The project successfully delivered:

- Automated customer journey analysis
- Funnel drop-off identification
- Traffic source evaluation
- Device performance analysis
- AI-generated business insights
- Live cloud deployment

The final solution transforms raw behavioral data into actionable recommendations for business stakeholders.

Future Improvements Scope

- Real-time analytics pipeline
- Advanced AI recommendation engine